

Unequal Job Opportunities and Well-Being Inequality: A Latent-Jobs Structural Decomposition

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Abstract

We study how unequal job opportunities contribute to inequality in money-metric well-being. The money metric is derived from the Measure-1 reference-set principle. Under the current empirical specification its direct reference collapses to the universally available non-employment state; opportunity heterogeneity therefore affects the current welfare measure through attained bundles. We model labour supply as a choice among latent jobs on French EU-SILC data, with EUROMOD computing taxes, benefits and disposable income at every alternative work arrangement. Preferences and opportunity components are jointly estimated, with their separation relying on maintained functional-form and exclusion restrictions. Attributing the resulting inequality to preferences and circumstances in full requires a counterfactual-attainment estimand that is still under design; ahead of it, we introduce and implement the bounded, preliminary three-factor structural exercise described below. In a preliminary three-factor structural exercise that holds household resources, needs and composition fixed, equalising systematic utility heterogeneity, coarse geographic/temporal access heterogeneity and earning opportunities changes money-metric well-being inequality by 1.8–9.9

Preliminary results throughout. The three-factor systematic-utility/geographic-temporal-access/earnings decomposition in Section 5 is a bounded exercise ahead of the paper’s final decomposition architecture, which requires a counterfactual-attainment estimand still under design; it is reported with Monte Carlo simulation ranges and an independent second-seed check, never confidence intervals, and household resources, needs and composition are held fixed rather than decomposed.

1 Introduction

Two people can work the same hours for the same hourly pay and be very differently placed. One chose that job from several that were open; the other took the only offer available. Their earnings are identical and their circumstances are not. Income comparisons record the outcome and cannot separate the two cases, and the difference matters for how unequal we should judge well-being to be.

The same ambiguity runs through the whole distribution. Low earnings can describe someone who values leisure highly and chose short hours, someone who cannot obtain a well-paid job, or someone whose family resources make a different work choice affordable. These are observationally entangled and economically different. A distribution of income is a distribution of outcomes with the explanations already mixed in.

The ambiguity matters for measurement, not only for interpretation. If we want to know how much of the inequality we observe is associated with unequal access to employment, hours arrangements and wage offers, we need two things that an income distribution does not supply.

We need a behavioural model that separates what a household wanted from what it could reach, and we need a rule for comparing people whose opportunities and whose tastes both differ. Neither is optional: without the first there is nothing to attribute, and without the second there is no defensible way to say who is better off.

The normative reference. Comparing well-being when individuals face different job opportunities requires a reference that specifies how those opportunities enter the comparison. We draw on the own-set equal-consumption criterion of Haydar and Maniquet (2026), work in progress. In its deterministic formulation, the criterion assigns to an attained situation the consumption level that would make the individual's preferred job within their own ability set equally good, when every feasible job provides that same consumption. The reference therefore retains the individual's own opportunities and own preferences while removing variation in pay from the reference bundles. Writing A_i for the jobs available to i and z_i for the attained situation, the reference level solves

$$u_i(z_i) = \max_{j \in A_i} u_i(W_i, j).$$

The argument being solved for is an amount of consumption, so the measure is money-metric by construction; there is no second conversion from an index into euros.

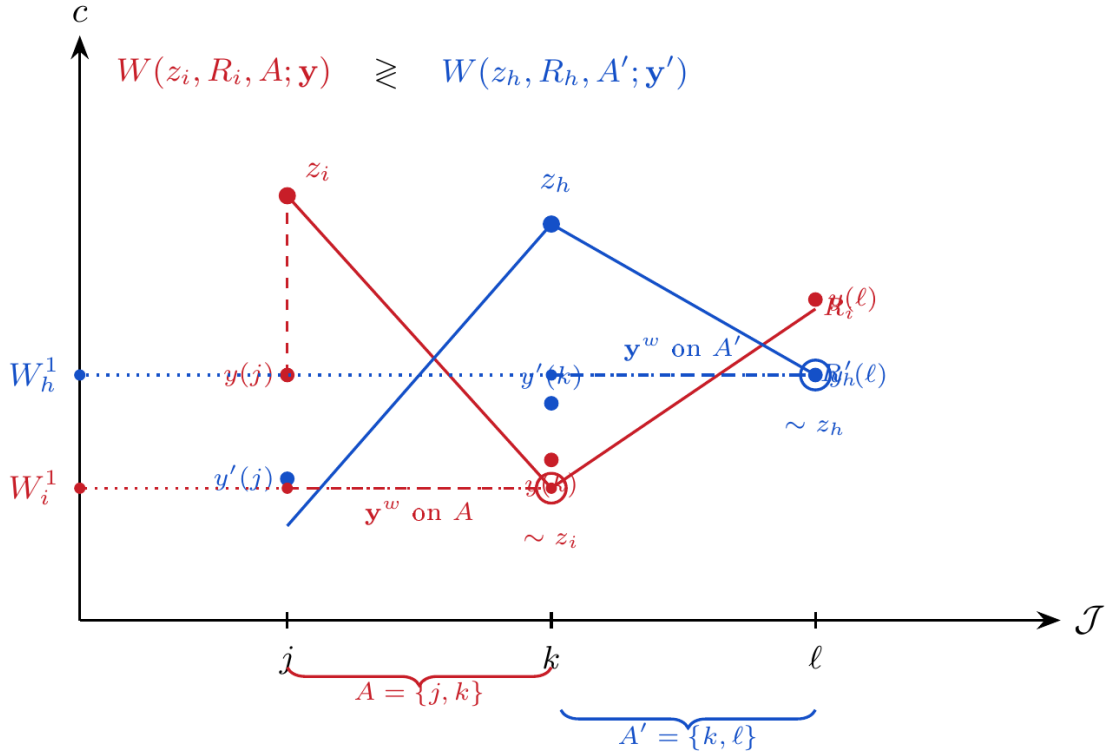


Figure 1: Own-set equal-consumption equivalents. The theoretical W1 construction from the companion theory paper. Individuals with preferences R_i, R_h and ability sets $A = \{j, k\}$ and $A' = \{k, \ell\}$ attain z_i and z_h . For each individual, a common consumption level is assigned to every job in their own set; the level at which the preferred reference job becomes indifferent to the attained bundle is the money metric. Adapted from Haydar and Maniquet (2026), work in progress. This is the deterministic construction; the estimated measure is its ex-ante extension, defined in Section 4.

The first figure depicts the theoretical W1 construction adapted from the companion theory paper by Haydar and Maniquet (2026). It constructs a deterministic own-set equal-consumption equivalent, using each individual's preferred reference job. Section 4 states the empirical implementation on the current estimated model – Mapping F, an accepted simplification that holds

because staying at home maximises the non-consumption index for every household in both samples – and derives its closed form; the deterministic figure does not depict an estimated welfare result.

One comparative static is worth stating plainly at the outset, because it is easy to misread. Holding the attained situation fixed, a person with a larger reference menu may need less uniform consumption to reach the same satisfaction. When actual opportunities change, however, both the attainment and the reference menu change. The net effect on equivalent consumption depends on both. Measuring attainment against one’s own opportunities is therefore not a claim that larger menus are intrinsically better, and we make no such claim.

The empirical approach. The behavioural half is a random-utility, random-opportunity model of job choice. A job is a package: an employment state, an occupation, a weekly hours arrangement and an hourly wage. Households rank packages by consumption and leisure, and the packages differ in how available they are. Preferences and opportunity components are jointly estimated, with their separation relying on maintained functional-form and exclusion restrictions. Neither is observed as a complete schedule. Every package is priced through the French tax-benefit system, so a change of hours or occupation moves disposable income through the actual schedule of taxes and transfers rather than a linear approximation. We estimate the model separately on 1,540 single-adult and 2,223 couple households drawn from French EU-SILC, with couples choosing jointly under a shared household budget.

The normative half then yields, for each household, an equivalent flat consumption level. Attributing its inequality to preferences and to circumstances in full requires a counterfactual-attainment estimand for the bundle a household would attain under a counterfactual environment, and that estimand is still under design. Ahead of it we report a bounded, preliminary exercise: we define three structural equalization operators, one for systematic utility heterogeneity, one for coarse geographic/temporal access heterogeneity and one for earning opportunities, holding household resources, needs and composition fixed throughout; we recompute every household’s welfare level, and the inequality of the resulting distribution, under each of the eight coalitions of those three operators, reusing the model’s own already-priced estimation panel with no re-estimation and no new pricing; and we allocate the interactions among them with an exact three-player Shapley value. The allocation is exhaustive by construction and the closure is verified numerically rather than imposed.

What we find. In a preliminary three-factor structural exercise that holds household resources, needs and composition fixed, equalising systematic utility heterogeneity, coarse geographic/temporal access heterogeneity and earning opportunities changes money-metric well-being inequality by 1.8–9.9% of the baseline Gini, depending on household type and reporting scale. Within the Shapley allocation of that movable component, earning-opportunity heterogeneity has a larger contribution than the coarse geographic/temporal access channel in both samples and both reporting conventions. The preference contribution is not sign-robust to equalisation. These are preliminary model-based accounting results, not causal estimates and not the final decomposition of total well-being inequality.

Dispersion in consumption is quantitatively large relative to dispersion in log well-being: $\text{Var}(\log C)$ is roughly 100–127% of $\text{Var}(\log W)$, with the excess offset by a large negative covariance between consumption and the leisure valuation. DECOMP-2 holds household resources, needs and composition fixed, leaving important sources of dispersion outside the P/A/B allocation.

Two qualifications belong with the headline. First, the preference contribution’s sign is not robust between the unequivalised and equivalised reporting conventions, in either sample: we therefore make no directional claim about it, only about the ordering of access against earnings. Second, an allocated share is an average of marginal contributions over coalition orders. It is not the reduction that equalizing that factor alone would achieve; Section 5 reports both and explains why they can differ, and reports the exact scale at which excluding the

model’s anchored-attainment arm moves ΔI , as a robustness check on the design rather than on the estimates. This is a bounded, preliminary reading, reported ahead of the paper’s final decomposition architecture, and it should be read as such throughout.

Relation to existing work. Every ingredient here has antecedents, and the contribution is the combination and the empirical answer, not any one step.

Modelling labour supply as choice among latent jobs, with preferences and opportunity components jointly estimated under maintained functional-form and exclusion restrictions, is due to [Aaberge et al. \(1995\)](#) and [Aaberge et al. \(1999\)](#), developed by [Dagsvik and Strøm \(2006\)](#) and [Dagsvik and Jia \(2016\)](#), surveyed by [Aaberge and Colombino \(2018\)](#), and set out for applied work by [Capéau et al. \(2016\)](#). We inherit that framework, including its distinction between the intensity of an offer and the probability of a choice, and its reliance on excluded opportunity shifters to separate components that choices alone do not separate. [Capéau et al. \(2016\)](#) also supply two of the identifying restrictions we maintain: the wage-offer distribution is independent of offered hours, and a local unemployment measure shifts availability while being excluded from preferences. Their Belgian application already covers single women, single men and couples, so covering both household types is not itself a contribution.

Combining such a model with a monetary welfare evaluation is also established. [Aaberge et al. \(1995\)](#) translate attained utility into an equivalent income against a stated reference household, choice set and tax system, and [Aaberge et al. \(2004\)](#) evaluate reforms with joint household choice and restricted hours. [Jia and Thoresen \(2021\)](#) make the case for the job-choice model in exactly this role and report a distribution of compensating variation for a Norwegian reform, and [Jacquet et al. \(2026\)](#) compare a standard compensating variation with one computed after replacing preference characteristics by reference values. All four evaluate a *change* between policy regimes. Our object is a cross-sectional distribution of levels under one policy system, and its attribution. A reform gain and a level have different origins and different denominators; a small preference-related reform difference would not imply a small preference share in our decomposition, and our shares do not estimate their reform gains.

The normative half draws on the literature that refuses to resolve interpersonal comparison by assuming common preferences. [Fleurbaey and Maniquet \(2006\)](#), [Fleurbaey and Maniquet \(2017\)](#) and [Fleurbaey and Maniquet \(2018\)](#) set out how a reference bundle encodes a position on compensation and responsibility; [Decoster and Haan \(2015\)](#) and [Bargain et al. \(2013\)](#) show empirically how much a welfare ordering moves with that choice. Their references fix a wage or an unearned-income intercept and maximize over a deterministic budget set. Ours fixes consumption across the alternatives of an estimated opportunity distribution and integrates. That is a different reference and, as Section 6 records, the ordering is sensitive to it.

The closest decomposition precedents are two. [Mühlhan \(2023\)](#) combines a structural labour-supply model with involuntary-unemployment restrictions and a Shapley attribution, decomposing the change in German household *income* inequality into temporal factors and binary restrictions. [Creedy and Hérault \(2011\)](#), in the section that constructs money-metric distributions under alternative policy and population states, decompose a change in inequality and social welfare by averaging over the two orders in which policy and population can be changed; theirs is the closest money-metric welfare-inequality decomposition we know. Both are decompositions of a *change* between two situations, with policy, population or temporal factors as the factors. Ours is a decomposition of a cross-sectional *level* of well-being inequality, with the factors defined as structural operators inside an estimated job-choice model: what a household prefers, which jobs it can reach, what those jobs pay, and what its budget and needs are. The wider behavioural-simulation decomposition tradition, [Bargain \(2012\)](#) and [Jessen \(2019\)](#), decomposes income changes in the same spirit.

The allocation rule is inherited outright. [Shorrocks \(1982\)](#) states the accounting requirements a decomposition should satisfy; [Shorrocks \(2013\)](#) places the Shapley value at the centre of distributional decomposition; [Sastre and Trannoy \(2002\)](#) documents how much the answer

can depend on how the exercise is set up. The preliminary three-operator exercise of Section 5 needs no grouping and uses the plain Shapley value on a three-player game. We claim no new allocation principle. Complete recomputation of every coalition, rather than a linear approximation, likewise has precedents and is better described as methodological discipline than as a contribution.

What is new, to our knowledge, is the conjunction: a random-utility random-opportunity model of job choice, an opportunity-sensitive money-metric welfare level built on the own-set equal-consumption principle, and a bounded, preliminary structural game over systematic utility heterogeneity, coarse geographic/temporal access heterogeneity and earning opportunities, with complete recomputation of welfare and inequality under every coalition. The application-specific methodological contribution is the game and its counterfactual operators, reported as preliminary and work in progress pending the counterfactual-attainment estimand of Section 4. Within the bounded DECOMP-2 game, earning-opportunity heterogeneity has a larger Shapley contribution than the model’s coarse geographic/temporal access channel. The cooperative-game allocation rule is inherited.

Roadmap. Section 2 describes the data and the household budget construction. Section 3 presents the latent-jobs model, its identifying restrictions and its estimation. Section 4 defines the money-metric welfare measure and the structural decomposition. Section 5 reports the behavioural and welfare results for both household types. Section 6 examines sensitivity and states the limitations. Section 7 concludes.

2 Data

The data are French EU-SILC, collection year 2016, with income reference year 2015. Taxes, benefits and disposable income are computed by EUROMOD ([Sutherland and Figari, 2013](#)) under the French 2015 policy system, which is the system in force over the income reference period. We use the same three dates consistently: the survey is collected in 2016, incomes refer to 2015, and the simulated policy system is 2015.

2.1 The estimation samples

Table 1: Sample construction. Unweighted households remaining after each screen, from the France 2016 EUROMOD input file to the two estimation samples. Rows are sequential; the counts are read from the frame records and are not reconstructed by subtraction.

Screen	Single-adult	Couple
the raw France 2016 input file	4,038	5,965
one- or two-adult households (composition screen)	4,038	5,965
every adult aged 20 to 60	2,131	3,662
no adult still in education	2,036	3,521
no old-age, disability or survivor pension receipt	1,755	3,218
labour status in scope	1,598	2,412
no other employable or earning member	1,564	2,323
observed hours and wage inside the calibrated support	1,555	2,275
hours outside [5,70] or unsupported military occupation (ISCO 0) on an employed decider	1,543	2,223

Screen	Single-adult	Couple
observed chosen alternative priced at non-positive disposable consumption	1,540	2,223
Estimation sample	1,540	2,223

The screens are of three kinds and it is worth separating them. The first two define the decision unit: the household must contain either one unpartnered adult or two mutually linked adults of opposite sex, and every decider must be between twenty and sixty. This is the largest exclusion and it is structural. Multi-generational households, flat-shares, adult children living with parents and same-sex couples are outside the estimated population, and no result extends to them. Because the age screen binds on both spouses, it removes proportionally more couples than singles.

The second kind removes households for whom the model does not define an offer set: adults in full-time education, households receiving an old-age, disability or survivor pension, and deciders whose labour-market status lies outside employment, unemployment and inactivity. Together with the pension screen this is why the surviving employment rate is high. The employment share below should be read as a share among people for whom working is a live option, not as a French employment rate.

The third kind is support. An employed decider's observed hours must lie in the closed interval $[5, 70]$ hours per week and the delivered hourly wage in $[2, 590]$ euros per hour, because those are the boundaries of the supports the opportunity densities are defined on. An observed occupation must map into the four modelled groups; ISCO 0, the armed forces, does not, and the resulting exclusion is *unsupported* occupation, not missing occupation. Three single-adult households whose own observed job prices to non-positive disposable consumption are removed, because the log-consumption term is undefined at their observed choice; non-positive *simulated* alternatives receive a one-euro consumption floor before utility is evaluated. Those are sampled alternatives entering the estimated choice likelihood. They do not enter the welfare measure of Section 4, which evaluates only each household's own observed job and its home state, both already screened to strictly positive consumption.

Table 2: The four occupation groups. ISCO-08 major groups are aggregated into four modelled groups; group 1 is the omitted reference in the estimated occupation block. This is a research aggregation adopted for this paper, not an ILO classification. ISCO 0, the armed forces, has no modelled alternative and is a sample screen; that is unsupported occupation, not missing occupation.

Model group	ISCO-08 major groups	Description
1 (reference)	6–9	Skilled agricultural, craft, plant and machine operators, and elementary occupations
2	5	Service and sales workers
3	4	Clerical support workers
4	1–3	Managers, professionals, technicians and associate professionals

2.2 What the households look like

Table 3: Descriptive means on the estimation samples. Weighted by the household cross-sectional weight; one row per household, with spouse-specific variables carried on the household row. Hours and wages are conditional on employment. These are observed inputs, not model predictions.

Measure	Single-adult decider	Couple man	Couple woman
Age (years)	41.1	39.8	37.8
Children under 20	—	—	—
Usual weekly hours, employed	37.6	41.1	35.8
Delivered hourly wage, employed (EUR/hour)	—	—	—
Households (unweighted)	1,540	2,223	

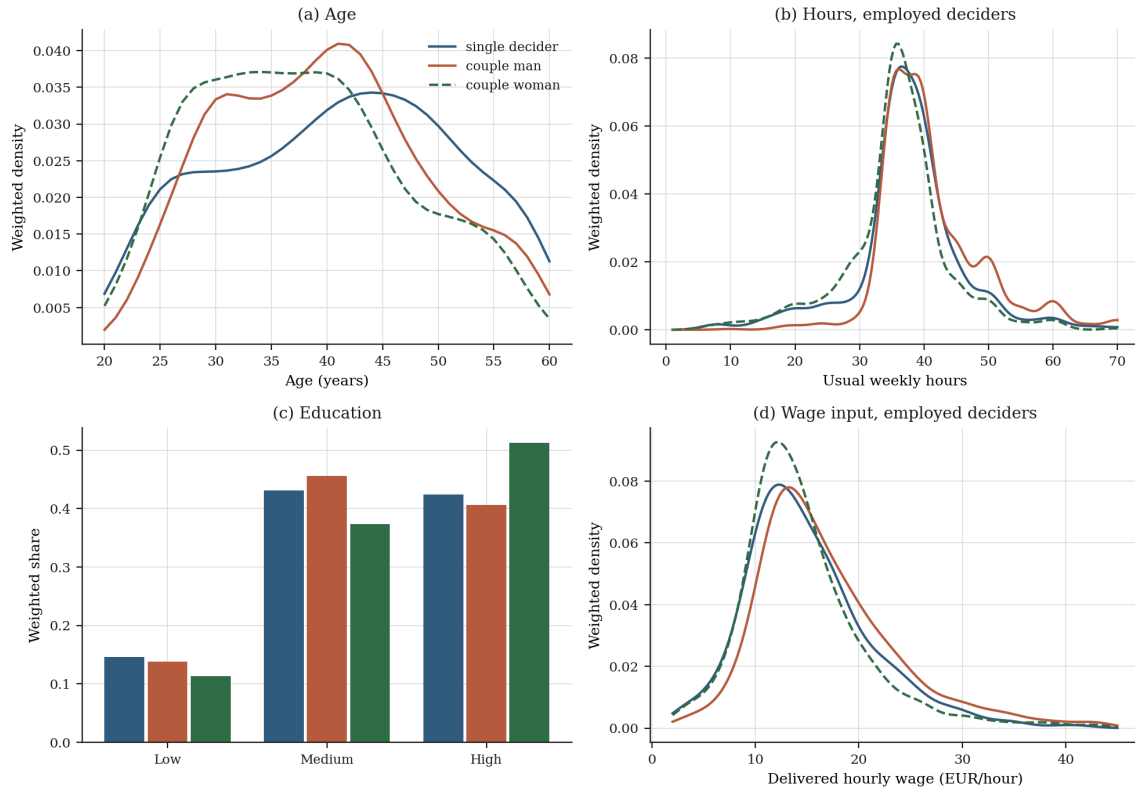


Figure 2: Weighted distributions on the estimation samples: age, usual weekly hours among employed deciders, education, and the delivered hourly wage input. Observed inputs, not model predictions.

Education is the three-level ISCED grouping, with the medium level as the omitted reference in the wage equation. Potential experience is years since leaving education, entering the wage equation in units of 20 years, with the square recomputed after scaling rather than carried forward. A child is any resident household member under 20, counted without a parent link. Age enters preferences centred on the sample mean decider age and divided by 10 years.

Table 4: Observed joint participation regimes among the 2,223 estimated couple households. These are joint regimes, not two spouse marginals.

Regime	Meaning	Weighted share	Mean hours, man	Mean hours, woman
BB	both spouses employed	0.8435	41.0	35.9
MO	man employed, woman not employed	0.0796	41.5	0.0
WO	woman employed, man not employed	0.0547	0.0	35.0
NN	neither spouse employed	0.0222	0.0	0.0

2.3 The household budget

For a given work arrangement the tax-benefit model is given the implied gross labour inputs, weekly hours, hourly wage, and earnings split at the thirty-five hour threshold, together with the household's non-labour inputs and roster, and returns disposable income. Only deciders receive counterfactual overrides; other members keep their baseline values. The accounting identity linking original income, benefits, taxes and social contributions to disposable income holds to machine precision at both the person and the household-alternative level, before and after the benefit take-up adjustment. Disposable income is summed over all resident members of the household.

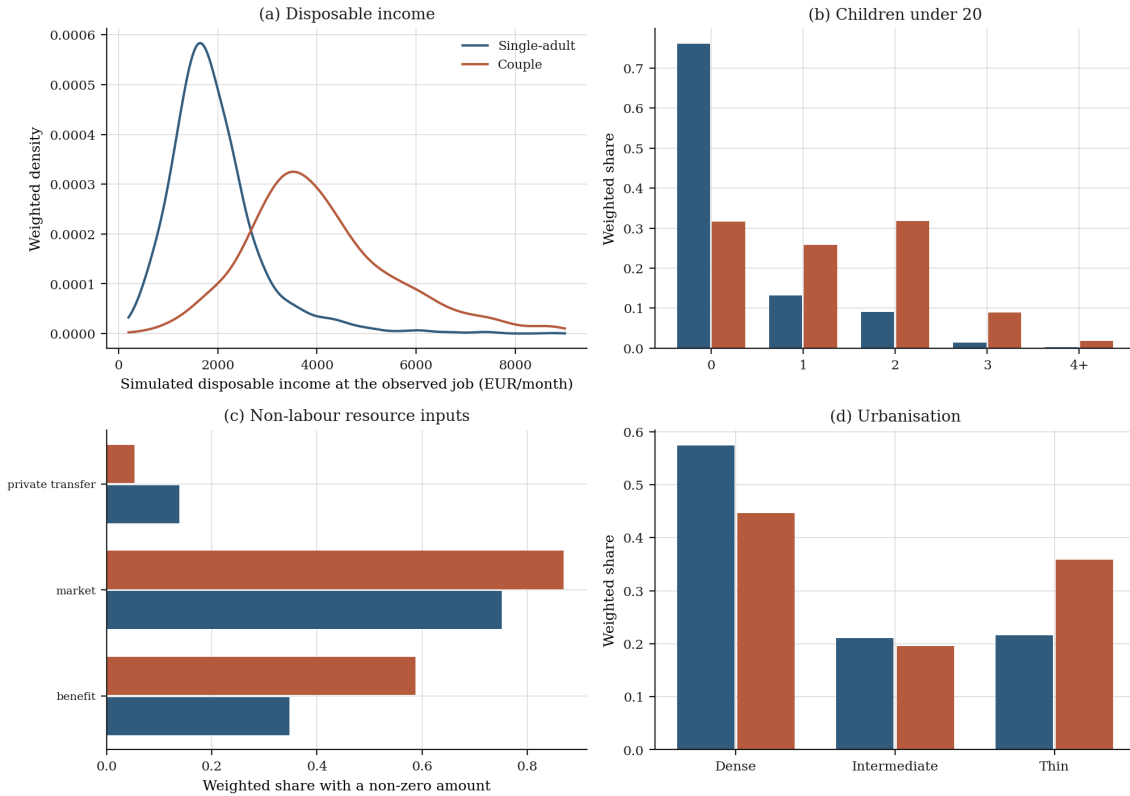


Figure 3: Weighted distributions on the estimation samples: simulated disposable income at the observed job, children under twenty, the incidence of non-labour budget inputs by family, and urbanisation. Panel (a) is a tax-benefit output evaluated at the observed choice; panel (c) reports inputs to that calculation. The two are never added, and stocks are never summed with monthly flows.

Panel (c) of the figure reports the *incidence* of non-labour budget inputs by family, not a cash total.

Some of these are stocks, some are annual flows and some are monthly; they are inputs to the tax-benefit calculation and are not added together into one income concept. The pension family is empty by construction, because pension receipt is a sample screen. Panel (a) reports the tax-benefit *output* at the observed job, which is an object of a different kind and is never added to panel (c).

Local labour-market conditions enter through a group unemployment measure, a population-weighted exposure index for the decider’s region and age-education cell, stored as a fraction and entering the employment index multiplied by 10. Regional and urbanisation variation supports the access specification conditional on the exclusion restriction and the functional form; it does not by itself identify access, and no causal effect of geography is claimed anywhere in this paper.

3 A latent-jobs model of household labour supply

A job is a package j : an employment state $e \in \{0, 1\}$ and, when employed, an occupation $k \in \{1, \dots, 4\}$, weekly hours $h \in [5, 70]$ and an hourly wage $w \in [2, 590]$. The base measure ν places counting mass on the single non-employment point and, on the employed part, counting measure over occupations with Lebesgue measure over hours and wages. A household chooses one package; a couple chooses one pair of packages under a shared budget.

3.1 Preferences

Let $C_i(j)$ be priced monthly disposable consumption at j and $\ell_i(j)$ normalized leisure, $(80 - h)/10$ hours per week, floored at one hour. For a single adult of sex group g , deterministic utility is

$$u_{ij} = \underbrace{\beta_\ell^g(\mathbf{x}_i) \mathcal{B}(\tilde{\ell}_{ij}; \theta_\ell^g)}_{L_{ij}} + \beta_c \log \tilde{c}_{ij}, \quad \beta_\ell^g(\mathbf{x}_i) = \beta_{\ell 0}^g + \beta_{\ell a}^g a_i + \beta_{\ell a^2}^g a_i^2 + \mathbb{1}\{g = \text{women}\} \beta_{\ell k}^g k_i,$$

where $\mathcal{B}(z; \theta) = (z^\theta - 1)/\theta$ with $\mathcal{B}(z; 0) = \log z$, $\tilde{\ell}_{ij}$ is normalized leisure, $\tilde{c}_{ij} = C_{ij}/\lambda_c$, a_i is centred age in decades and k_i is the child count. For a couple the index is additive across spouses with no direct cross-leisure term,

$$u_{ij} = \beta_\ell^m(\mathbf{x}_i) \mathcal{B}(\tilde{\ell}_{ij}^m; \theta_\ell^m) + \beta_\ell^f(\mathbf{x}_i) \mathcal{B}(\tilde{\ell}_{ij}^f; \theta_\ell^f) + \beta_c \log \tilde{c}_{ij},$$

with consumption the tax-unit sum over all members. We write $L_i(j)$ throughout for the complete non-consumption part of the index, which for a couple contains both leisure terms.

The systematic leisure specification is not age alone: it carries an intercept, age, age squared and group-specific leisure curvature for each of the four adult groups, plus a number-of-children shifter for women; sex and household type are carried by separate parameter blocks. “The baseline deliberately keeps systematic preference heterogeneity parsimonious: age profiles for all four adult groups and a child-related shifter for women. The child term is interpreted as a reduced-form behavioural/time-constraint shifter, not as pure taste.”

Three features of this specification are decisions and are worth naming. The consumption curvature is exactly zero, so the consumption term is $\beta_c \log(C/\lambda_c)$ and the marginal utility of consumption is β_c/C , independent of the normalizer. The consumption weight β_c is **estimated**, not fixed at one. And the random-utility shock is i.i.d. type-I extreme value with scale one, so utility is measured in the natural unit of that scale. Fixing the shock scale is a normalization; additionally fixing β_c would be a substantive restriction relative to that normalization, and estimating β_c removes it. Neither step identifies an absolute cardinal utility scale, and fixing the consumption functional form is not by itself what identifies the shock scale.

The normalizer λ_c is a units convention. Under exact log consumption it enters the index as the alternative-invariant constant $-\beta_c \log \lambda_c$, so it cancels from every choice probability; it does not appear in the accepted welfare measure of Section 4 at all.

3.2 Opportunities

The opportunity density gives the relative intensity with which packages are available to household i , before any choice is made. It is a product of **four** factors, switched off on non-employment by the employment indicator $E_{ij} = \mathbb{1}\{h_{ij} > 0\}$:

$$g_{ij} = g_{ij}^E \cdot (g_{ij}^H \cdot g_{ij}^{\text{Occ}} \cdot g_{ij}^W)^{E_{ij}}.$$

Access, $g_{ij}^E = \exp\{\beta_E + \beta_s s_i + \sum_{r=2}^8 \beta_r \text{reg}_{ir} + \beta_u u_i + \beta_m m_i\}$, carries the local group unemployment measure s_i , the region indicators and urbanisation, and is the factor inside which local-market access lives. **Hours**, $g_{ij}^H = \exp\{\sum_b \beta_b \mathbb{1}\{h_{ij} \in b\}\}$, elevates five bands over a residual reference set of total width 26.5 hours per week, so every band coefficient is read against that residual and none of the five is normalized to zero; the narrow full-time band $[33.5, 36.5)$ is a density elevation over an interval, not an atom at thirty-five hours. **Occupation**, $g_{ij}^{\text{Occ}} = \exp\{\beta_{k,g}^{\text{occ}}\}$ with $\beta_{1,g}^{\text{occ}} \equiv 0$. **Wage offer**, g_{ij}^W , is log-normal with location $\mu_i = \beta_{w0} + \beta_{wL} L_i + \beta_{wH} H_i + \beta_{wx} x_i + \beta_{wx^2} x_i^2 + \delta_{\text{occ}}$ and dispersion σ , truncated to $[2, 590]$ euros per hour and renormalized on that support.

Two properties of the normalization matter later, for estimation and for the identification argument below. Write $\hat{g}_{ij} = g_{ij}/Z_i$ for the density normalized to unit mass on the whole package space. The wage factor integrates to one *conditional on employment and occupation*, because the truncated density is renormalized on its own support; and the non-consumption index L_{ij} has no wage argument. The pay-neutrality property of the welfare reference established in Section 4 rests on a simpler fact than either of these: the reference job itself, staying at home, carries no opportunity-density argument at all, so none of g , the proposal, the shock or an intensity parameter enters the reference.

The full opportunity model contains employment access, hours, occupation and wage-offer blocks. DECOMP-2 does not equalise all of them: its A channel is limited to local geographic/temporal access shifters (region, urban/rural, year), while personal occupation access, hours-band access and other alternative characteristics remain fixed; B carries earning opportunities.

3.3 Identification, in words

Choices alone do not separate a taste for leisure from a scarcity of jobs at that number of hours. Three kinds of restriction do the work. Preferences are smooth in hours through the leisure index, while the opportunity density is a step function over bands, so a spike in observed hours at a band is read as availability rather than as a kink in tastes. Excluded shifters enter availability and not preferences: the local group unemployment measure and the regional and urbanisation indicators shift the employment index and appear nowhere in utility. And the wage-offer distribution is specified as independent of offered hours conditional on occupation, which separates the wage location from the hours density. These are the restrictions of Capéau et al. (2016) and Dagsvik and Jia (2016), maintained here rather than tested. The regional variation supports the access block conditional on those restrictions; it does not establish separate identification on its own, and it is not a causal design.

Preferences and opportunity components are jointly estimated, with their separation relying on maintained functional-form and exclusion restrictions. No causal interpretation is attached to that separation.

3.4 Estimation

The choice set is a continuum, so the likelihood is evaluated over sampled alternatives. For each household we draw $R = 100$ alternatives from a proposal density q_{ij} and place the observed package in the set as well, giving a set $\mathcal{C}_i$ with $|\mathcal{C}_i| = 101$ rows. Write the sampled-set index

$$V_{ij} = u_{ij} + \log g_{ij} - \log q_{ij}, \quad q_{ij} = q_{ij}^E (q_{ij}^H q_{ij}^W q_{ij}^{\text{Occ}})^{E_{ij}}.$$

The contribution of household i is then the conditional probability of its observed package y within that set,

$$\Pr(y | \mathcal{C}_i) = \frac{n_y \exp V_{iy}}{\sum_{s \in \mathcal{C}_i} \exp V_{is}}.$$

The denominator runs over slots rather than distinct packages, and a package drawn more than once keeps its multiplicity, n_y ; the observed package carries the same proposal correction as any other row. The proposal density is a computational device and never an economic object: it is not an offer, an availability or an opportunity. The proposal is fitted out of fold, so a household's own outcome never enters the proposal used to score it. For couples the proposal draws the joint participation regime first and then the two spouse packages conditional on it.

Standard errors are cluster-robust on the household. Optimization uses five starts under two polishing contracts, ten terminal paths in all; we report the spread of the criterion across those paths and the

eigenvalues of the exact Hessian at the selected optimum. Those diagnostics support a stable local solution found from the starts tested. They are not a proof of global uniqueness, and we do not claim one.

Two features of the observation rule remain genuinely open and are stated here rather than in a footnote. The sample screens on the observed hours and wage of employed deciders, so the estimation sample is selected on an outcome of the process being modelled. The conditional likelihood above is the correct object for a sampled choice set; it is not automatically the correct object for an outcome-selected sample, and we have not established which correction, if any, the screen requires. Separately, the derivation of the sampled-set probability conditions on a labelled collection of slots; small observed cross-coordinate correlations and a binomial-looking multiplicity distribution are consistent with independent draws but do not by themselves establish the sampling law. Both are recorded as unresolved in Section 6.

4 Money-metric well-being and structural inequality decomposition

4.1 The verified Mapping-F W_F^1 construction

The accepted welfare measure (Deputy ruling R1, Mapping F, `JMP_W1_fork_ruling_v1.md`) is the amount of consumption that, paid at a fixed reference job o , leaves the household exactly as well off as at its attained situation:

$$W_{i,F}^1 = m_i(o; z_i), \quad u_i(m_i(o), o) = u_i(z_i).$$

This is the theoretical own-set equal-consumption criterion above (Measure 1), $u_i(z_i) = \max_{j \in A_i} u_i(W_i, j)$, specialised to the reference job o that the maximisation over the household's own reachable set actually selects. On the current empirical domain the selected reference job is always staying at home: for every household in both samples, non-employment maximises the non-consumption index, so the maximisation collapses to a single evaluation at the household's own observed job against its own home state, o . Under the log-consumption specification of Section 3 this gives the closed form

$$W_{i,F}^1 = C_i^{\text{obs}} \exp \left[\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right],$$

where C_i^{obs} is the household's own priced disposable consumption at its observed job, $L_i(j_i^{\text{obs}})$ and $L_i(o)$ are the non-consumption index at the observed job and at the home state, and β_c is the estimated consumption weight. Four objects do not enter this construction at all: the opportunity density $\hat{g}$, the numerical proposal q , the behavioural shock, and any latent-set intensity parameter – opportunities act on this measure only through the attained bundle j_i^{obs} , never through an average or an integral over jobs the household did not take. That home always maximises the non-consumption index, so that W^1 coincides here with the staying-home equivalent, is an empirical-domain property of this specification and this estimated model, verified directly rather than assumed; it is not a theorem of the Haydar-Maniquet framework, which defines the reference job o as whichever job the maximisation selects and does not guarantee it is staying at home.

The construction is independently verified. A reconstruction that does not import the production code reproduces every pre-registered check – non-workers equal their own observed consumption exactly, workers strictly below it, no route by which $\hat{g}$, q , the shock, or an intensity parameter can enter – and reproduces the committed sample aggregates for both populations to machine precision (`baseline_f1_verification_v1.md`, MNL commit 6048c9f7, independently verified b5550af5).

4.2 What the reference does and does not do to pay

The money metric is derived from the Measure-1 reference-set principle. Under the current empirical specification its direct reference collapses to the universally available non-employment state; opportunity heterogeneity therefore affects the current welfare measure through attained bundles.

The reference does not depend directly on pay. The accepted construction's home reference (Mapping F, Deputy ruling R1) is a fixed singleton state whose systematic utility carries no opportunity-density, proposal, behavioural-shock or intensity argument, so none of those four objects enters it at all.

Changes in earning opportunities nevertheless change the measure, because they change what the household attains, not the reference it is compared against. The correct statement is therefore narrow and we make only it: *the reference is directly pay-neutral, and earning opportunities reach the measure through the attained evaluation*. We do not report a magnitude for that attained-side channel here: quantifying it under a counterfactual equalization of earning opportunities is a decomposition question, and Section 5 reports only the bounded, preliminary exercise available for that. A qualitative channel existing does not establish that the deterministic independence-of-pay axiom fails for this stochastic functional. That would require fixing the primitives the axiom holds fixed and proving the property, which we have not done and do not claim.

4.2.1 A worked illustration, in the current construction

Not a description of any real household. For a one-nat shortfall, $L_i(j_i^{\text{obs}}) - L_i(o) = -1$, the closed form gives $W_{i,F}^1 = C_i^{\text{obs}} \exp(-1/\beta_c) < C_i^{\text{obs}}$. Current nonworkers: $W = C$. Current workers: $W < C$ under the maintained empirical domain. This is a mechanical illustration of the closed form, not an empirical claim about any particular household.

4.3 Interpersonal comparison and the unit

The measure is defined at the household. Comparing households of different size requires an equivalence scale, which is a normative choice and not an estimate. We report every result on two bases: a raw household basis, and an equivalized basis using the modified OECD scale. We never pool the two household types into one distribution, because the two applications carry different reference constructions and the levels are not comparable; every share below is a share of the baseline inequality of its own population.

4.4 The preliminary structural decomposition

A complete attribution of measured inequality to preferences and to circumstances requires a counterfactual-attainment estimand: under a counterfactual environment, which bundle does the household attain? Two candidate estimands are under design – a realised-bundle route conditioning the behavioural latent state on the observed choice, and an ex-ante route integrating the measure over the model-implied counterfactual choice distribution – and neither is executed here, because the inequality of expected welfare is not the expected inequality of welfare and the two routes answer different questions. Ahead of that design choice, this section reports a bounded, explicitly preliminary exercise that reuses the model’s own already-priced estimation panel, with no re-estimation and no new pricing, and simulates each household’s attained bundle under a counterfactual environment by carrying the household’s realised draws through it directly.

Let $X_i = (P_i, A_i, B_i)$ collect three structural inputs for household i :

- P_i systematic utility heterogeneity: the leisure-weight covariates and the sex- or spouse-specific preference block, including the reduced-form time-constraint shifters this pathway also carries;
- A_i local geographic/temporal access shifters (region, urban/rural, year): only those arguments of the employment index; personal occupation access, hours-band access, the group-unemployment measure and node-level alternative characteristics remain fixed;
- B_i earning opportunities: the covariates entering the offered-wage location.

Household resources, needs and composition are held fixed throughout this exercise rather than treated as a fourth operator: they are not equalized in any coalition and no share is attributed to them. Let $\mathcal{I}$ be the Gini index and $W_i(\cdot)$ the money metric of the previous subsection, so the baseline is $I_\emptyset = \mathcal{I}\{W_i(X_i)\}_{i=1}^N$. For each factor define a **structural equalization operator** T_P, T_A, T_B , which replaces that factor’s arguments across all households by a common reference profile and leaves the estimated coefficients in place. For a coalition $S \subseteq \{P, A, B\}$ let $X^S = T_S(X)$ be the state in which exactly the factors in S are equalized, and set

$$I_S = \mathcal{I}\{W_i(T_S X)\}_{i=1}^N, \quad v(S) = I_\emptyset - I_S.$$

v is a cooperative game on three players: the worth of a coalition is the inequality it removes when its factors are equalized together. T_S is a single simultaneous substitution map, not an ordered product $\prod_{k \in S} T_k$, and an operator changes a *pathway*, not every occurrence of a raw characteristic.

Table 5: The three structural equalization operators of the preliminary decomposition (Section 5). Each operator replaces the arguments of one structural pathway with a common reference profile and leaves the estimated coefficients in place; it does not equalize every occurrence of a raw characteristic. Operators are applied as one simultaneous substitution map, not as an ordered product. Household resources, needs and composition are held fixed throughout this exercise – not a fourth operator here – pending the counterfactual-attainment estimand the paper’s final decomposition architecture requires.

Operator	What is replaced	What is retained
T_P preferences (systematic utility heterogeneity)	Singles: the arguments of the leisure weight and the complete reference-sex leisure block. Couples: the medoid spouse arguments, with own spouse coefficients retained	Access and wage pathways of the same characteristics
T_A local geographic/temporal access shifters	Region, urban/rural and year arguments of the employment index	Preferences, wage location, group unemployment, hours-band access, personal occupation access and node-level alternative characteristics
T_B earning opportunities	The arguments of the offered-wage location: education shares and experience moments, with squares recomputed rather than averaged	The estimated wage coefficients and dispersion; the preference and access pathways of the same characteristics

Welfare and inequality are then recomputed from the model for all eight coalitions of $\{P, A, B\}$. There is no linearisation and no re-estimation: coefficients are held at their estimates throughout, and only the arguments move.

The interactions are allocated with the **exact Shapley value** (Shorrocks, 2013) on the three-player game v : the average of each factor’s marginal contribution over all orderings in which the three factors can be equalized. With three players this average has a closed form over $3! = 6$ orderings and requires no grouping. The allocation is exhaustive: the contributions sum to $\Delta I = I_\emptyset - I_{\{P,A,B\}}$, and we report ΔI itself rather than treat it as identically zero, because household resources, needs and composition remain in the residual $I_{\{P,A,B\}}$ along with sex-block parameter differences and behavioural randomness.

Uncertainty is reported two ways, never merged into a confidence interval: a Monte Carlo range across 1,000 simulation replications at fixed estimates, and an independent check under a second simulation seed. Neither is a parameter-uncertainty interval; propagating the estimation covariance into this exercise is future work.

5 Empirical results

5.1 Behavioural estimates

The estimated model has 41 free coordinates for single adults, of which 40 are interior, and 47 for couples, all interior. The criterion is 6253.463 and 10283.034 respectively. Across ten terminal paths from five starts under two polishing contracts, the criterion varies by 1.1e-10 for single adults and 1.3e-09 for couples, and the smallest eigenvalue of the exact Hessian on the interior block is 0.1669 and 0.0771. One single-adult coordinate, the female age-square term in the leisure weight, sits at a box endpoint; its interval is reported under the active-set convention in the appendix and it is excluded from the interior curvature.

Table 6: The preference block. Cluster-robust standard errors in parentheses, clustered on the household. “At bound” marks an estimate at a box endpoint, whose interval is reported under the active-set convention in the appendix. The consumption weight is common within a population and enters as $\beta_c \log(C/\lambda_c)$.

Coefficient	Single men	Single women	Couple men	Couple women
Leisure weight, intercept $\beta_{\ell 0}$	8.5199 (3.5480)	5.8683 (2.1346)	3.9814 (0.7466)	13.1029 (4.7760)
Leisure weight, age $\beta_{\ell a}$	1.4820 (1.0236)	0.0678 (0.4792)	-0.0080 (0.0287)	-0.1228 (0.1260)
Leisure weight, age squared $\beta_{\ell a^2}$	0.7783 (0.7732)	1.0000 (at bound)	0.0065 (0.0030)	0.0073 (0.0103)
Leisure weight, children $\beta_{\ell n}$	—	0.1666 (0.4422)	—	-0.3055 (1.1472)
Leisure curvature θ_ℓ	-1.6263 (0.3301)	-0.9274 (0.2161)	-0.9761 (0.1357)	-1.6863 (0.2458)
Consumption weight β_c	2.0387 (0.2917)	2.0387 (0.2917)	2.1017 (0.2939)	2.1017 (0.2939)

The consumption weight is 2.0387 with a cluster-robust standard error of 0.2917 for single adults and 2.1017 with 0.2939 for couples. Two things follow. In the closed-form welfare measure of Section 4, β_c is the rate at which a household’s non-consumption advantage at its observed job over staying home, $L_i(j_i^{\text{obs}}) - L_i(o)$, converts into a proportional adjustment of its own observed consumption: the same advantage moves the money metric proportionally less, the larger β_c is, because the advantage enters the exponent divided by β_c . And because utility is on the unit-scale shock, one natural unit of the index corresponds to multiplying consumption by $\exp(1/\beta_c)$, a factor of 1.633 for single adults and 1.609 for couples. That is a proportional statement: there is no single euro value of a unit of utility independent of the consumption at which it is evaluated.

Table 7: The opportunity block: employment access, the hours density and occupation access. Cluster-robust standard errors in parentheses. The employment index multiplies the working indicator; the omitted references are region 1, thinly populated areas, the residual hours set of total width 26.5 hours per week, and occupation group 1. The singles occupation column reports the female coordinates; the male coordinates are in the appendix table. Hours coefficients are common across the two singles sex groups and spouse-specific for couples.

Coefficient	Singles	Couple men	Couple women
Employment constant β_E	-3.1735 (0.4053)	-2.1233 (0.3195)	-2.8148 (0.3021)
Group unemployment rate $\beta_{E,gsur}$	-1.4422 (0.2356)	-1.1924 (0.1557)	shared with the man
Densely populated $\beta_{E,u}$	-0.0288 (0.2209)	-0.1823 (0.1657)	shared with the man
Intermediate density $\beta_{E,m}$	0.0641 (0.2614)	-0.4174 (0.1861)	shared with the man
Region 2 $\beta_{E,2}$	-0.3780 (0.3292)	-0.1546 (0.2442)	shared with the man
Region 3 $\beta_{E,3}$	-0.1117 (0.3891)	0.0873 (0.2808)	shared with the man
Region 4 $\beta_{E,4}$	-0.8218 (0.3810)	0.0150 (0.3068)	shared with the man
Region 5 $\beta_{E,5}$	-0.5162 (0.3274)	-0.1455 (0.2555)	shared with the man
Region 6 $\beta_{E,6}$	-0.7222 (0.3528)	-0.2844 (0.2748)	shared with the man
Region 7 $\beta_{E,7}$	-0.5374 (0.3503)	-0.1396 (0.2651)	shared with the man
Region 8 $\beta_{E,8}$	-0.4512 (0.3401)	-0.1659 (0.2660)	shared with the man
Hours band: Part-time lower, [17.5, 21.5)	0.0298 (0.1972)	-1.1094 (0.3791)	-0.5173 (0.1629)
Hours band: Part-time upper, [28.5, 30.5)	0.7080 (0.1928)	0.0835 (0.2378)	1.1586 (0.1214)
Hours band: Narrow full-time, [33.5, 36.5)	2.0656 (0.0894)	2.2888 (0.0821)	2.0284 (0.0682)

Coefficient	Singles	Couple men	Couple women
Hours band: Full-time upper, [36.5, 40.5]	1.9387 (0.0997)	2.3801 (0.0866)	1.6825 (0.0798)
Hours band: Long hours, [44.5, 70]	-0.0834 (0.1772)	0.6977 (0.1335)	-0.3109 (0.1615)
Occupation 2 ξ_2	0.1135 (0.1374)	-1.5097 (0.0967)	0.2074 (0.0891)
Occupation 3 ξ_3	-0.4060 (0.1440)	-2.2532 (0.1244)	-0.1804 (0.0923)
Occupation 4 ξ_4	0.5632 (0.1225)	0.1594 (0.0618)	0.8159 (0.0799)

Table 8: The wage block and the consumption weight. The offered log wage is normal with mean $\mu_i(k)$ and dispersion σ , truncated to [2, 590] EUR/hour and renormalized on that support. Experience enters in units of twenty years. Slopes and σ are common across the two singles sex groups and across spouses within couples; the singles and couples models are estimated separately, so the two columns are not restricted to agree.

Coefficient	Singles	Couples
Intercept β_{w0}	2.0138 (0.0561)	2.0480 (0.0335)
Low education β_{wL}	0.0566 (0.0372)	-0.0433 (0.0217)
High education β_{wH}	0.1491 (0.0308)	0.1817 (0.0197)
Potential experience β_{wx}	0.2408 (0.0862)	0.5643 (0.0546)
Experience squared β_{wx^2}	-0.0282 (0.0390)	-0.1598 (0.0243)
Occupation 2 δ_2	-0.0347 (0.0399)	-0.0740 (0.0231)
Occupation 3 δ_3	0.0622 (0.0383)	0.0394 (0.0228)
Occupation 4 δ_4	0.2785 (0.0368)	0.2146 (0.0223)
Log-wage dispersion σ	0.3815 (0.0133)	0.3631 (0.0072)
Consumption weight β_c	2.0387 (0.2917)	2.1017 (0.2939)

Maintained restrictions. Singles: `theta_c_singles = 0`; the couples preference block (`beta_l0_m`, `beta_l_age_m`, `beta_l_age2_m`, `beta_l0_f`, `beta_l_age_f`, `beta_l_age2_f`, `beta_l_nkids_f`, `theta_l_f`) does not enter the singles likelihood; and `beta_E_y2015` and `beta_E_y2017` are absent because the 2015 and 2017 data are not used. Couples: `theta_c = 0`, the male children leisure effect is structurally zero, and the direct cross-leisure term is fixed at zero. These are restrictions, not estimates. The singles and couples models are estimated separately, so their wage blocks are not restricted to agree, and the difference between the two experience profiles should not be read as a test.

Indifference curves in consumption and leisure, single-adult households

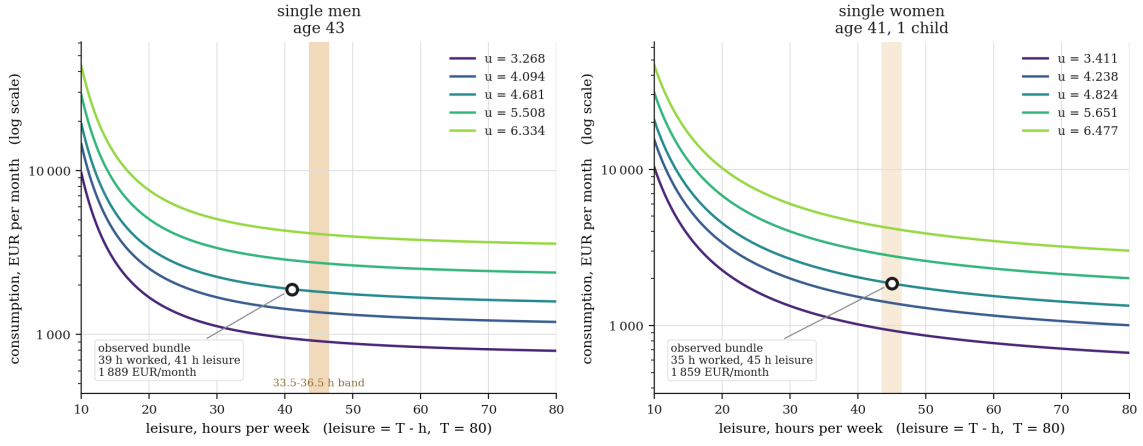


Figure 4: Indifference curves in consumption and leisure, single-adult households, at one representative household of each sex. The budget set, the opportunity density and the taste shock are not drawn, so a curve is not a set of attainable bundles. The consumption axis is logarithmic; under log consumption every finite utility target is attainable at a finite positive consumption, which may lie outside the plotted range.

Indifference curves in consumption and leisure, couple households, one panel per spouse

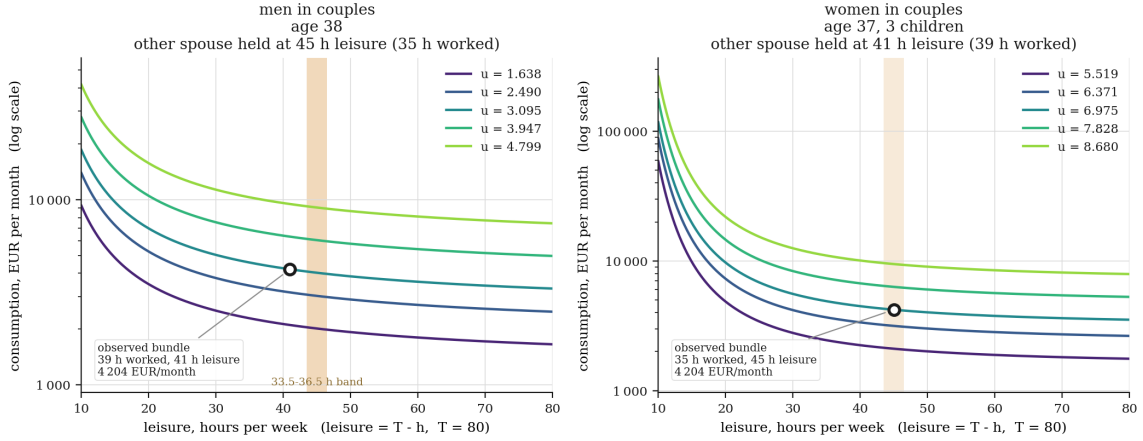


Figure 5: Indifference curves in consumption and each spouse's leisure, couple households, with the partner's leisure held at its observed value. Conditional slices of a joint object, not attainable sets.

The local compensation u_ℓ/u_c by age and sex, and the exact compensation for one more weekly hour

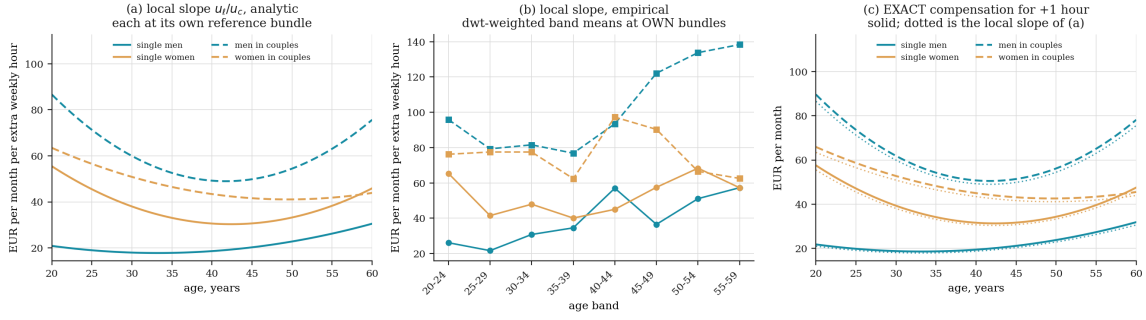


Figure 6: Marginal rate of substitution between leisure and consumption, by age and sex, in euros per month per additional recurring weekly hour of leisure. A compensation along an indifference curve, not a behavioural response to a wage change.

The indifference curves and the marginal rate of substitution are the economics; the coefficients are coordinates. Note that the consumption axis is logarithmic and the curves are drawn over the plotted range only: under $L(\ell) + \beta_c \log(C/\lambda_c)$, any finite utility target at positive leisure is reached at the finite positive consumption $C = \lambda_c \exp\{(\bar{u} - L(\ell))/\beta_c\}$, which is strictly positive for every finite target. A curve leaving the frame has left the plotting range, not the domain of the model, and nothing here is economically infeasible.

5.2 Fit

The fit reported here is a population prediction, computed by integrating the estimated model over the opportunity distribution and the taste shocks. It is not a sampled-menu choice probability and it is not an in-sample fitted value. We report it band-referenced: each group's weighted extensive-margin accuracy is compared not to a 100 per cent target – a correctly specified stochastic-choice model does not attain one – but to the 95 per cent band that model itself would produce by chance, from 500 outcome vectors simulated at the fitted estimates.

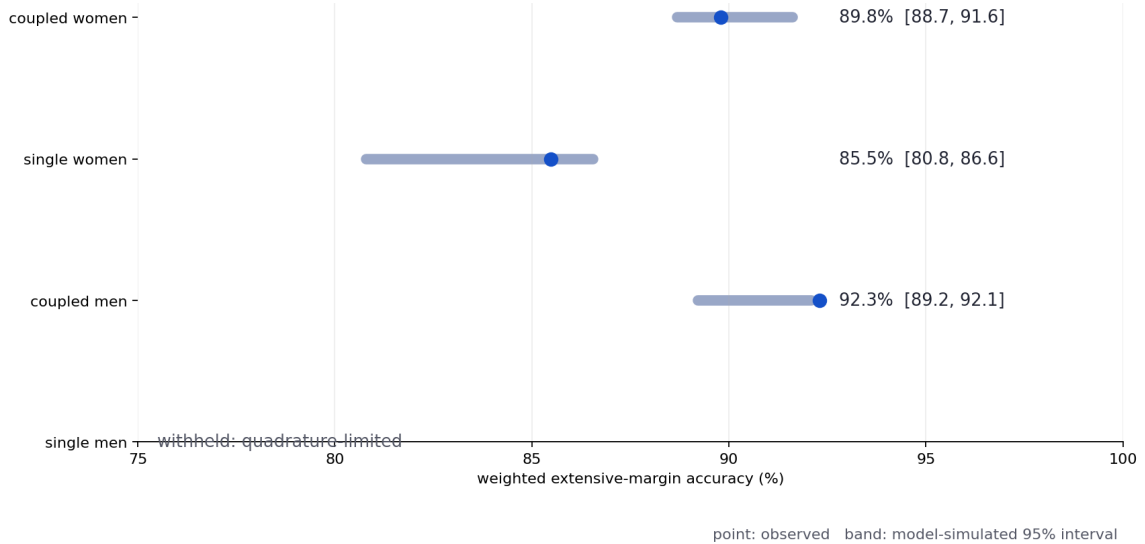


Figure 7: Weighted extensive-margin accuracy against the model's own simulated 95 per cent band (500 outcome vectors at the fitted estimates), restricted to groups whose statistic clears the pre-registered numerical-adequacy gate; single men do not clear it and are withheld. Observed and band values: coupled women 89.8% [88.7, 91.6]; single women 85.5% [80.8, 86.6]; coupled men 92.3% [89.2, 92.1]. Source: POSFIT v3 (MNL_posfit, branch diagnostics/posfit-v3, commit 96693269).

Aggregate participation and occupation margins are reproduced reasonably closely, while hours fit is

uneven. Individual-level predictive diagnostics raise possible excess stochastic dispersion in some groups; the group-level classification is not used as a headline result.

The figure below shows each group’s weighted extensive-margin accuracy against its own simulated 95 per cent band, restricted to groups whose extensive-accuracy statistic clears the pre-registered numerical-adequacy gate (Monte Carlo simulation error at most 0.25 times the same sampling variation): coupled women, coupled men and single women clear it on the weighted statistic and are shown; single men do not and are withheld rather than shown with a caveat, because a statistic the quadrature cannot resolve is not a weaker finding, it is not a finding. Coupled men clear the weighted gate only narrowly – ratio 0.240 against the 0.25 threshold – and miss the unweighted version, 0.250; the weighted rule is applied throughout. Among the displayed groups, observed accuracy is inside its band for coupled women and single women, and above the top of its band for coupled men. Clearing this extensive-accuracy gate is a narrower question than the group-level fit classification discussed in the limitations below, which draws on a wider set of statistics and is inconclusive – not passing – for single women specifically. Section 6 records where the fuller diagnostic set – calibration by predicted decile, the joint hours/occupation confusion structure, and the group withheld here – still has open questions.

5.3 Within-sample equivalised results

Table 9: The verified Mapping-F W_F^1 construction, modified-OECD equivalised, against equivalised disposable consumption in the same sample. Household EUR/month, dwt-weighted. Source: BASELINE-F-1 (MNL 6048c9f7, independently verified b5550af5) and its equivalised reporting (commit 4c4e07e). Reported separately by population; no pooled figure and no cross-population level comparison.

PopulationN		Workers	Non-workers	C^{eq} mean	C^{eq} median	C^{eq} Gini	W_F^{1eq} mean	W_F^{1eq} median	W_F^{1eq} Gini
Single-adult	1,540	1,336	204	1,766	1,588	0.2633	1,302	1,165	0.2498
Couple	2,223	2,173	50	2,245	2,063	0.2268	1,311	1,237	0.1974

The verified W_F^1 construction gives, on the modified-OECD-equivalised basis, a weighted Gini of 0.2498 for single-adult households and 0.1974 for couples, against 0.2633 and 0.2268 for equivalised disposable consumption in the same samples. This is a within-sample descriptive dispersion comparison, not a welfare-loss statement: the ratio W_F^1/C^{obs} reflects the systematic leisure term at the observed job relative to home leisure, and we do not construct a loss index from it. Singles and couples are reported separately throughout, with no pooled distribution and no cross-sample level comparison: the two are separate estimation frames with different welfare units, and a comparison of the two levels is not drawn here or elsewhere in this paper.

5.4 The preliminary decomposition

Table 10: Single adults, the eight P/A/B coalitions. Weighted Gini of money-metric well-being, dwt-weighted, simulated on the accepted model’s already-priced estimation panel with no re-estimation and no new pricing. “Change from actual” is the one-factor effect of that coalition. The Monte Carlo range is the spread of the Gini level across 1,000 simulation replications, never a confidence interval.

Scale	Coalition	I(S), Gini	Change from actual	MC range (min–max)
unequalised	Actual (no equalization)	0.2337	+0.0000	0.2203–0.2509
unequalised	P	0.2326	-0.0011	0.2180–0.2491
unequalised	A	0.2319	-0.0018	0.2178–0.2480
unequalised	B	0.2298	-0.0039	0.2148–0.2452
unequalised	P + A	0.2310	-0.0027	0.2177–0.2478
unequalised	P + B	0.2282	-0.0054	0.2138–0.2437
unequalised	A + B	0.2282	-0.0055	0.2135–0.2411

Scale	Coalition	I(S), Gini	Change from actual	MC range (min-max)
unequalised	P + A + B	0.2268	-0.0069	0.2127-0.2431
equalised	Actual (no equalization)	0.2451	+0.0000	0.2284-0.2663
equalised	P	0.2456	+0.0005	0.2304-0.2647
equalised	A	0.2436	-0.0014	0.2264-0.2631
equalised	B	0.2419	-0.0032	0.2264-0.2592
equalised	P + A	0.2443	-0.0008	0.2304-0.2624
equalised	P + B	0.2418	-0.0033	0.2263-0.2594
equalised	A + B	0.2405	-0.0046	0.2245-0.2582
equalised	P + A + B	0.2406	-0.0045	0.2248-0.2580

Table 11: Couples, the eight P/A/B coalitions. Weighted Gini of money-metric well-being, dwt-weighted, simulated on the accepted model’s already-priced estimation panel with no re-estimation and no new pricing. “Change from actual” is the one-factor effect of that coalition. The Monte Carlo range is the spread of the Gini level across 1,000 simulation replications, never a confidence interval.

Scale	Coalition	I(S), Gini	Change from actual	MC range (min-max)
unequalised	Actual (no equalization)	0.2035	+0.0000	0.1936-0.2144
unequalised	P	0.1956	-0.0079	0.1858-0.2059
unequalised	A	0.2025	-0.0010	0.1938-0.2140
unequalised	B	0.1892	-0.0143	0.1803-0.2018
unequalised	P + A	0.1946	-0.0089	0.1844-0.2046
unequalised	P + B	0.1842	-0.0193	0.1742-0.1962
unequalised	A + B	0.1883	-0.0152	0.1796-0.2000
unequalised	P + A + B	0.1834	-0.0201	0.1731-0.1944
equalised	Actual (no equalization)	0.1971	+0.0000	0.1871-0.2075
equalised	P	0.1969	-0.0002	0.1873-0.2069
equalised	A	0.1963	-0.0008	0.1869-0.2055
equalised	B	0.1887	-0.0084	0.1789-0.1998
equalised	P + A	0.1961	-0.0010	0.1861-0.2049
equalised	P + B	0.1927	-0.0044	0.1829-0.2039
equalised	A + B	0.1880	-0.0091	0.1775-0.1985
equalised	P + A + B	0.1921	-0.0050	0.1821-0.2033

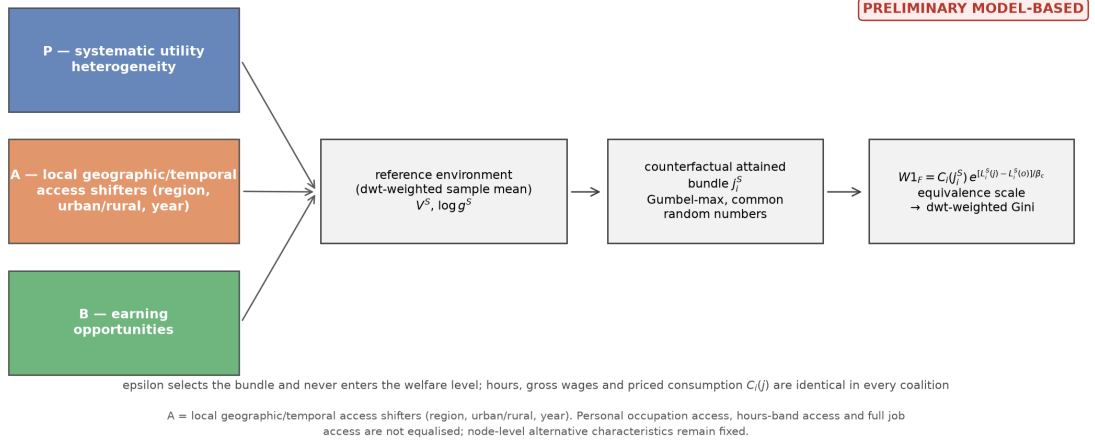


Figure 8: The eight P/A/B counterfactual coalitions, built from the accepted model by equalising household-constant covariates within a block. Node-level alternative characteristics are preserved in every coalition; only household-constant covariates are equalised. Preliminary, model-based.

The bounded percentage reported below is not evidence that opportunities are unimportant; it is the change inside the stated partial accounting game.

The decomposition is bounded by design because household resources, needs and composition are held fixed. Within that bounded game, equalising P/A/B changes the Gini by 1.8–9.9% of its baseline level, depending on sample and reporting scale.

Dispersion in consumption is quantitatively large relative to dispersion in log well-being: $\text{Var}(\log C)$ is roughly 100–127% of $\text{Var}(\log W)$, with the excess offset by a large negative covariance between consumption and the leisure valuation. DECOMP-2 holds household resources, needs and composition fixed, leaving important sources of dispersion outside the P/A/B allocation. Sex-block parameter differences and behavioural randomness also remain in the residual and are not attributed to P , A or B .

Table 12: The exact three-player Shapley allocation of ΔI across P , A and B . Gini-point contribution beside the share of ΔI (not of baseline inequality), and an independent second-seed reproduction. No directional claim is made about P : its sign changes between unequivalised and equivalised reporting in both samples, so its share of ΔI is not stated. A Monte Carlo per-replication share range, which divides by that replication’s own near-zero ΔI and is not informative on its own, is reported in the technical gallery, not here.

Population	Scale	Factor	Label	Gini-point contribution	Share of ΔI	Second-seed Gini-point
Single-adult	unequivalised	P	Preferences (systematic utility heterogeneity)	+0.0012	—	+0.0011
Single-adult	unequivalised	A	Local access shifters (region, urban/rural, year)	+0.0016	23.4%	+0.0017
Single-adult	unequivalised	B	Earning opportunities	+0.0040	58.6%	+0.0042
Single-adult	unequivalised	ΔI	$I(\text{actual}) - I(P,A,B \text{ equalized})$	+0.0069	100.0%	+0.0071

Population	Scale	Factor	Label	Gini-point contribution	Share of ΔI	Second-seed Gini-point
Single-adult	equivalised	P	Preferences (systematic utility het- erogeneity)	-0.0003	—	-0.0004
Single-adult	equivalised	A	Local access shifters (region, urban/rural, year)	+0.0013	29.6%	+0.0014
Single-adult	equivalised	B	Earning opportunities	+0.0034	77.4%	+0.0038
Single-adult	equivalised	ΔI	I(actual) – I(P,A,B equalized)	+0.0045	100.0%	+0.0049
Couple	unequivalised	P	Preferences (systematic utility het- erogeneity)	+0.0064	—	+0.0065
Couple	unequivalised	A	Local access shifters (region, urban/rural, year)	+0.0009	4.6%	+0.0009
Couple	unequivalised	B	Earning opportunities	+0.0128	63.5%	+0.0127
Couple	unequivalised	ΔI	I(actual) – I(P,A,B equalized)	+0.0201	100.0%	+0.0201
Couple	equivalised	P	Preferences (systematic utility het- erogeneity)	-0.0019	—	-0.0019
Couple	equivalised	A	Local access shifters (region, urban/rural, year)	+0.0007	14.6%	+0.0008
Couple	equivalised	B	Earning opportunities	+0.0062	124.0%	+0.0061
Couple	equivalised	ΔI	I(actual) – I(P,A,B equalized)	+0.0050	100.0%	+0.0050

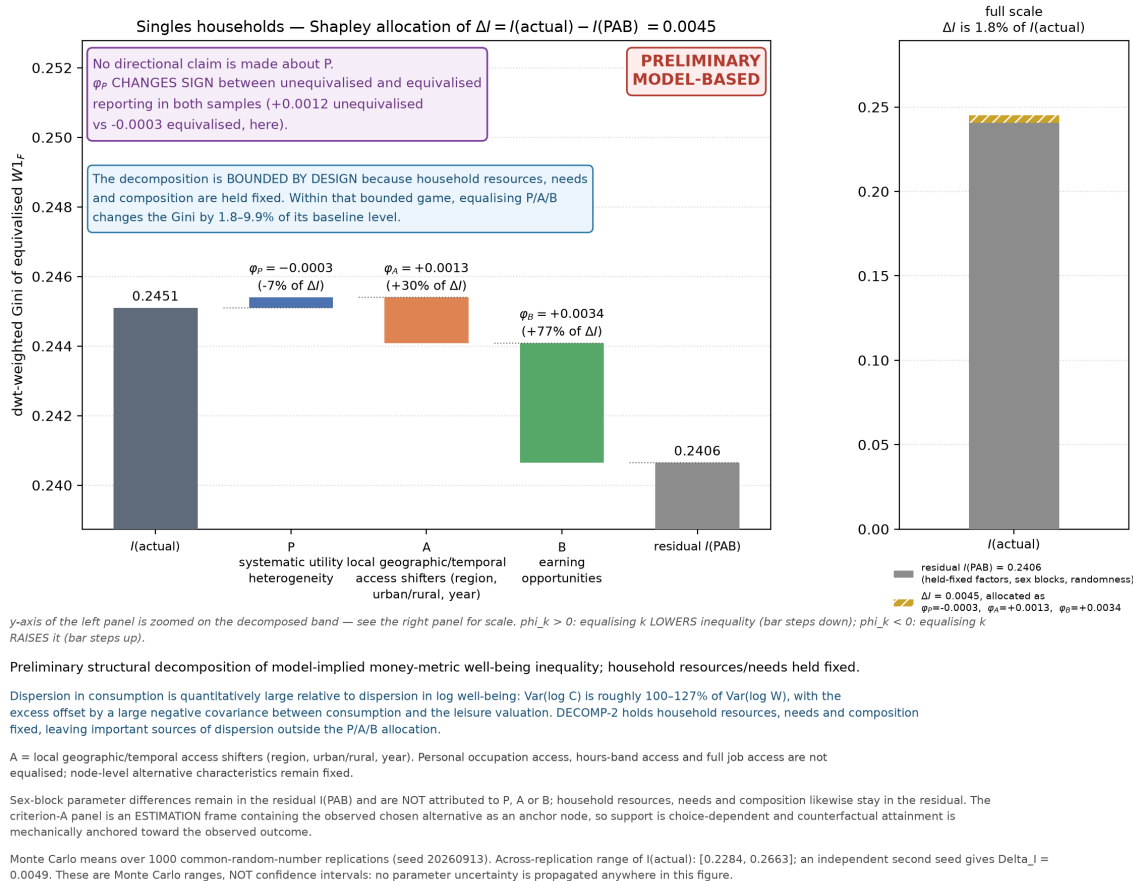


Figure 9: Single-adult estimation sample: coalition Gini levels and the exact Shapley allocation of ΔI across P, A and B, with the sign instability of P annotated. Modified-OECD-equivalised $W1_F$; Monte Carlo ranges over 1,000 replications, not confidence intervals. Preliminary, model-based.

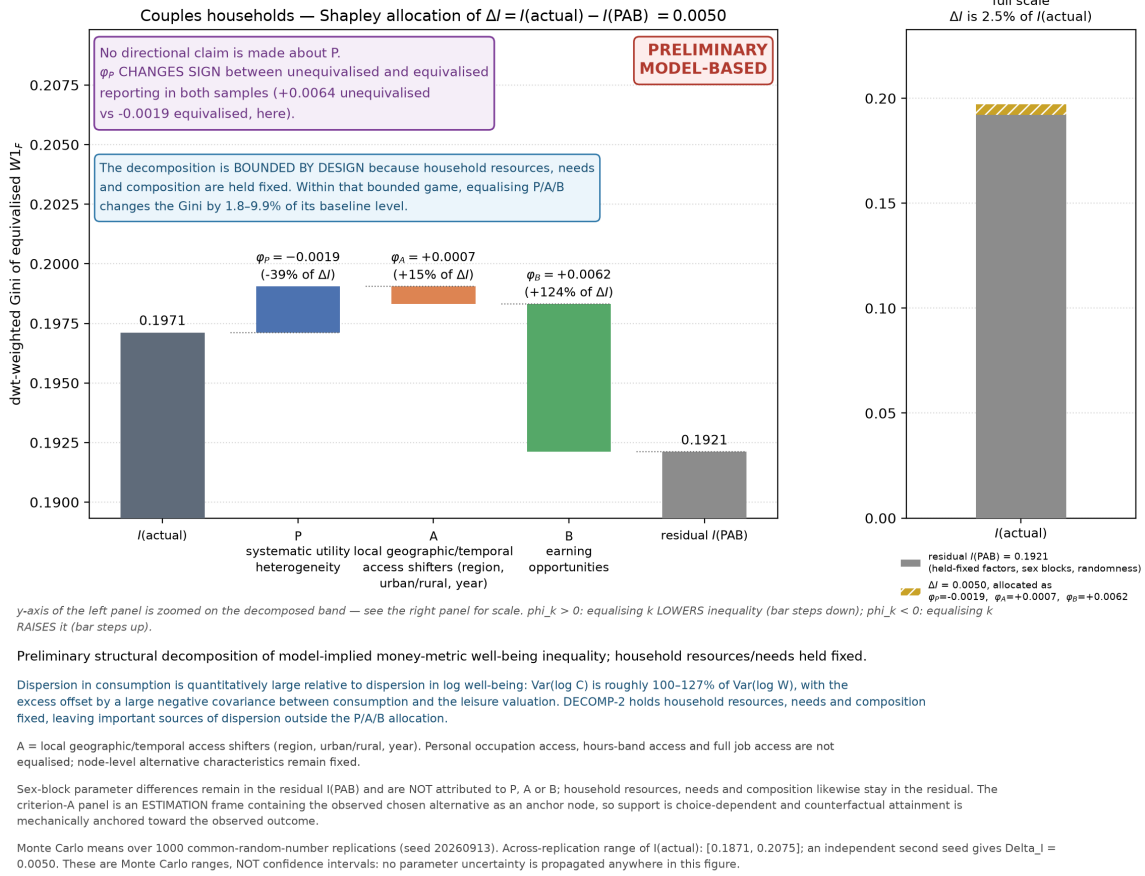


Figure 10: Couple estimation sample: coalition Gini levels and the exact Shapley allocation of ΔI across P, A and B, with the sign instability of P annotated. Modified-OECD-equivalised $W1_F$; Monte Carlo ranges over 1,000 replications, not confidence intervals. Preliminary, model-based.

In a preliminary three-factor structural exercise that holds household resources, needs and composition fixed, equalising systematic utility heterogeneity, coarse geographic/temporal access heterogeneity and earning opportunities changes money-metric well-being inequality by 1.8–9.9% of the baseline Gini, depending on household type and reporting scale. Within the Shapley allocation of that movable component, earning-opportunity heterogeneity has a larger contribution than the coarse geographic/temporal access channel in both samples and both reporting conventions. The preference contribution is not sign-robust to equivalisation.

These are preliminary model-based accounting results, not causal estimates and not the final decomposition of total well-being inequality.

The allocated shares are not one-factor effects. An allocated share is an average of marginal contributions over the orders in which coalitions can form; the coalition table above reports the one-factor effect directly, as the change from the actual coalition for the single-letter row. The two can differ, and with three players and an exact Shapley value the difference is fully accounted for by the interaction terms, which the closed form makes explicit.

Robustness. Every coalition Gini level and every Shapley contribution reproduces closely under an independent second simulation seed, including the sign instability of the preference contribution, which appears in both runs. Excluding the estimation panel's anchor node – the household's own observed choice, inserted deterministically in every coalition, which mechanically anchors counterfactual attainment toward the observed outcome to a small, roughly common degree (2.9 per cent of single adults and 3.9 per cent of couples attain it under any coalition) – moves ΔI by at most 9.3 per cent (couples, equivalised), with no sign flip in any specification.

The allocation is exhaustive to numerical precision: the residual after summing P, A, B and ΔI back to $I_\emptyset$ is zero to machine precision in every sample and scale, verified rather than imposed. That is a computational validation of the accounting for the declared game. It does not validate the identification

of the model, the normative content of the operators, or a resolution of the counterfactual-attainment question this exercise is deliberately bounded around.

6 Sensitivity and limitations

6.1 Equivalization and the preliminary decomposition

Equivalization is a normative choice, and the preliminary decomposition of Section 5 moves with it. On the equivalized basis the movable share ΔI is larger relative to baseline inequality for both household types than on the raw household basis, and the earning-opportunities/coarse-geographic-and-temporal-access ordering is unchanged in every case: earning opportunities have the larger contribution at both scales, for both household types. The preference contribution’s sign already differs between the two scales at the raw comparison reported in Section 5; equivalization is one of the two axes that sign instability spans, not an independent further concern. Both bases are reported throughout; neither is the correct one, and the choice between them is not settled by the data. We do not report a male-primary or other alternative reference-block variant of this preliminary exercise; that sensitivity, reported for the welfare baseline itself in Section 4, has not been re-run through the bounded decomposition.

6.2 Two open econometric questions

The sample screens on the observed hours and wage of employed deciders. The estimation sample is therefore selected on an outcome of the process being modelled, and the conditional sampled-set likelihood, which is the right object given a sampled choice set, is not automatically the right object given an outcome-selected sample. We have not established which correction the screen requires, or that none is required. This is an econometric question and not a presentational one, and it is not resolved by the diagnostics reported above.

Separately, the sampled-set probability is derived by conditioning on a labelled collection of slots. Small observed cross-coordinate correlations and a multiplicity distribution consistent with independent draws are diagnostics; they do not by themselves establish the sampling law that the derivation assumes. Both questions are recorded here as open.

6.3 Limitations

The limitations travel with the results above, named together here rather than scattered through the text they qualify.

The A channel is local geographic/temporal access only. The operator A_i of Section 4 carries region, urban/rural status and survey year; it does not carry any individual capability, education-specific or occupation-specific access channel, hours-band access or full job access. What the preliminary decomposition attributes to A is therefore the coarse geographic and temporal component of access, not access in the fuller sense of what an individual with particular skills or credentials can reach; a more personal access channel, if identified, would move some inequality currently attributed elsewhere into A , in a direction we do not sign.

ΔI is bounded by design. The decomposition is bounded by design because household resources, needs and composition are held fixed. Within that bounded game, equalising P/A/B changes the Gini by 1.8–9.9% of its baseline level, depending on sample and reporting scale. The restriction is by construction; the realised numerical magnitude is a model result under that restriction.

The preference contribution’s sign depends on the reporting scale. ϕ_P , the exact Shapley contribution of preferences, changes sign between the unequivalised and equivalised reporting conventions in both samples. We therefore make no directional claim about the preference contribution, only about the ordering of the coarse geographic/temporal access channel against earning opportunities, which is scale-robust.

Single men’s extensive-accuracy statistic is quadrature-limited. It fails the pre-registered numerical-adequacy gate and is withheld rather than reported at all (Section 5); the raw employment rate is matched closely (1.3 percentage points), but on the margin where this group’s fit is worst, the near-full-time hours band, the record specification under-predicts by 9.5 percentage points, the largest single discrepancy in either sample. This is a genuine, group-specific fit limitation, not a tail-coverage artefact of the kind reported for the sub-ten-hour cell.

Wage elasticities are omitted. A valid gross-wage perturbation requires new tax-benefit repricing over the affected job alternatives, and the current priced support does not contain that counterfactual. No approximate or mock elasticity figure is reported.

The time endowment, $T = 80$ hours a week, is a convention the model is not indifferent to. A diagnostic re-estimation at $T = 75$ and $T = 90$, holding the rest of the certified protocol fixed, moves the criterion materially in both directions and changes which coordinates are interior: at $T = 90$ a second leisure-curvature coefficient joins the boundary that is interior at $T = 80$. $T = 80$ is therefore a maintained convention, not a free normalization the results are invariant to, and it has not itself been estimated or justified.

Individual-level predictive diagnostics raise possible excess stochastic dispersion in some groups; the group-level classification is not used as a headline result. The pre-registered composite decision rule flags couples (both sexes) and single men, on the weighted statistics, with a misspecification-evidence verdict against the model’s own simulated variation, rather than a mechanical consequence of conditioning on outcomes. Single women’s verdict is inconclusive, but not because they fail the extensive-margin numerical-adequacy gate reported above – they clear that gate on both weighted and unweighted statistics. Their inconclusive verdict traces instead to a different set of statistics (the prediction-conditioned regression-to-the-mean check and several hard-classification and reliability statistics) failing their own, separately applied adequacy gate. Two of the four group verdicts are themselves sensitive to the weighting convention – coupled men’s underlying gate status and single men’s misspecification verdict do not survive switching from weighted to unweighted statistics. This is diagnostic evidence from a composite decision rule, not a single statistic, it is sensitive to unsupported finite-panel regions being floored rather than excluded, and the underlying verdicts remain open pending further review; we report it as an open question about the model’s implied stochasticity, not as an established fact about it.

6.4 What is not established

The preliminary decomposition is an accounting of a structural model under declared operators. It is not a causal analysis. No regional, educational or occupational effect reported here is identified as a causal effect, and the exhaustiveness of the allocation validates the accounting for the declared game, not the identification of the model or the normative content of the operators.

Section 5’s decomposition is bounded and preliminary in three specific ways, stated together here. It holds household resources, needs and composition fixed rather than decomposing them, leaving important sources of dispersion outside the P/A/B allocation; the reported 1.8–9.9% change is defined within that bounded game. It reuses the model’s already-priced estimation panel to simulate counterfactual attainment, which anchors that attainment toward the observed outcome to a small, checked degree, rather than resolving the counterfactual-attainment estimand the paper’s final decomposition architecture requires. And its uncertainty is reported as a Monte Carlo simulation range and an independent second-seed check, not as a parameter-uncertainty interval propagated from the estimation covariance. None of the three is a defect in what is reported; together they are why it is reported as preliminary rather than as the paper’s decomposition result.

7 Conclusion

Observed hours and earnings do not say whether a household chose its position or settled for it, and that ambiguity is not a nuisance for welfare measurement: it is the substance of it. This paper takes the ambiguity seriously in both halves of the problem. On the behavioural side, preferences and opportunity components are jointly estimated, with their separation relying on maintained functional-form and exclusion restrictions; every alternative is priced through the tax-benefit system. On the normative side, the money metric is derived from the Measure-1 reference-set principle. Under the current empirical specification its direct reference collapses to the universally available non-employment state; opportunity heterogeneity therefore affects the current welfare measure through attained bundles.

A complete decomposition of the resulting inequality into preferences and circumstances awaits a counterfactual-attainment estimand still under design. In a preliminary three-factor structural exercise that holds household resources, needs and composition fixed, equalising systematic utility heterogeneity, coarse geographic/temporal access heterogeneity and earning opportunities changes money-metric well-being inequality by 1.8–9.9% of the baseline Gini, depending on household type and reporting scale. Within the Shapley allocation of that movable component, earning-opportunity heterogeneity has a larger

contribution than the coarse geographic/temporal access channel in both samples and both reporting conventions. The preference contribution is not sign-robust to equivalisation.

These are preliminary model-based accounting results, not causal estimates and not the final decomposition of total well-being inequality.

Three limits should travel with that reading. The quantitative opportunity decomposition is preliminary and work in progress, bounded by construction, not the paper’s final decomposition. The allocation is exhaustive for the declared three-player game, which validates the accounting and not the identification or the ethics of the operators. And several questions named in Section 6 – about the observation rule, about the model’s own scope, and about where it under- or over-predicts – remain genuinely open, unrelated to the decomposition itself. What the exercise offers is not a causal account of why opportunities differ, and not yet a complete attribution of measured inequality, but a disciplined, preliminary statement of which of two circumstance channels dominates, once preferences and a structural model are made explicit and the reference is stated.

A Appendix A. The estimated parameter vectors

The tables below report only coordinates estimated on the reported sample. **Maintained restrictions.** Singles: $\theta_{c_singles} = 0$; the couples preference block (β_{10_m} , $\beta_{1_age_m}$, $\beta_{1_age2_m}$, β_{10_f} , $\beta_{1_age_f}$, $\beta_{1_age2_f}$, $\beta_{1_nkids_f}$, θ_{1_f}) does not enter the singles likelihood; and β_{E_y2015} and β_{E_y2017} are absent because the 2015 and 2017 data are not used. Couples: $\theta_{c_} = 0$, the male children leisure effect is structurally zero, and the direct cross-leisure term is fixed at zero. These are restrictions, not estimates.

One single-adult coordinate, the female age-square term, is at its box endpoint. Under the active-set convention its interval is not reported, the interior curvature is computed after removing it, and the parameter draws used for the welfare intervals hold it at its estimate.

Table 13: The singles estimated coordinates. Standard errors are cluster-robust on the household; boundary-active estimates are marked at bound. Maintained restrictions are stated once in the accompanying note.

Coordinate	Estimate	CR1 s.e.	z	Box	Status
β_{10_sm}	8.51993	3.54798	2.401	[0.05, 50]	estimated
$\beta_{1_age_sm}$	1.48195	1.02361	1.448	[-5, 5]	estimated
$\beta_{1_age2_sm}$	0.778284	0.773158	1.007	[-1, 1]	estimated
θ_{1_sm}	-1.62627	0.330058	-4.927	[-8, 0.95]	estimated
β_{10_sf}	5.86829	2.13459	2.749	[0.05, 50]	estimated
$\beta_{1_age_sf}$	0.0677662	0.479217	0.141	[-5, 5]	estimated
$\beta_{1_age2_sf}$	1	–	–	[-1, 1]	at bound
$\beta_{1_nkids_sf}$	0.166636	0.442236	0.377	[-5, 5]	estimated
θ_{1_sf}	-0.927353	0.216128	-4.291	[-8, 0.95]	estimated
$\beta_{E_}$	-3.17353	0.405337	-7.829	[-25, 25]	estimated
β_{h_pt1}	0.0297672	0.197247	0.151	[-10, 10]	estimated
β_{h_pt2}	0.707971	0.192788	3.672	[-10, 10]	estimated
β_{h_ft}	1.93873	0.0996968	19.446	[-10, 10]	estimated
β_{h_lh}	-0.0833752	0.177201	-0.471	[-10, 10]	estimated
β_{E_gsur}	-1.44224	0.235591	-6.122	[-10, 10]	estimated
β_{E_drgn2}	-0.377963	0.32925	-1.148	[-10, 10]	estimated
β_{E_drgn3}	-0.111713	0.38912	-0.287	[-10, 10]	estimated
β_{E_drgn4}	-0.821813	0.380963	-2.157	[-10, 10]	estimated
β_{E_drgn5}	-0.516158	0.327362	-1.577	[-10, 10]	estimated
β_{E_drgn6}	-0.722159	0.35278	-2.047	[-10, 10]	estimated
β_{E_drgn7}	-0.537369	0.350308	-1.534	[-10, 10]	estimated
β_{E_drgn8}	-0.45119	0.340074	-1.327	[-10, 10]	estimated
β_{E_drgur}	-0.0287689	0.220916	-0.130	[-10, 10]	estimated
β_{E_drgmd}	0.064133	0.261352	0.245	[-10, 10]	estimated
$\beta_{occ_2_m}$	-1.37371	0.158147	-8.686	[-15, 15]	estimated
$\beta_{occ_3_m}$	-2.10963	0.202248	-10.431	[-15, 15]	estimated

Coordinate	Estimate	CR1 s.e.	z	Box	Status
beta_occ_4_m	-0.325897	0.121341	-2.686	[-15, 15]	estimated
beta_occ_2_f	0.113492	0.137399	0.826	[-15, 15]	estimated
beta_occ_3_f	-0.406009	0.144045	-2.819	[-15, 15]	estimated
beta_occ_4_f	0.563216	0.122467	4.599	[-15, 15]	estimated
beta_w0	2.01382	0.0560922	35.902	[-10, 20]	estimated
beta_w_educL	0.0566261	0.0372485	1.520	[-5, 5]	estimated
beta_w_educH	0.149054	0.0308408	4.833	[-5, 5]	estimated
beta_w_pexp	0.24077	0.0862223	2.792	[-1, 1]	estimated
beta_w_pexp2	-0.0281739	0.0389905	-0.723	[-0.1, 0.1]	estimated
sigma	0.381505	0.0132752	28.738	[0.1, 20]	estimated
delta_occ_2	-0.0347155	0.039899	-0.870	[-4, 4]	estimated
delta_occ_3	0.0622139	0.0382749	1.625	[-4, 4]	estimated
delta_occ_4	0.278494	0.0367705	7.574	[-4, 4]	estimated
beta_h_f35	2.06556	0.0894479	23.092	[-10, 10]	estimated
beta_c	2.03873	0.291729	6.988	[0.05, 50]	estimated

Table 14: The couples estimated coordinates. Standard errors are cluster-robust on the household; boundary-active estimates are marked at bound. Maintained restrictions are stated once in the accompanying note.

Coordinate	Estimate	CR1 s.e.	z	Box	Status
beta_10_m	3.98136	0.746569	5.333	[1e-06, 50]	estimated
beta_1_age_m	-0.00799228	0.0287174	-0.278	[-5, 5]	estimated
beta_1_age2_m	0.00648336	0.00297388	2.180	[-1, 1]	estimated
theta_1_m	-0.976109	0.135663	-7.195	[-8, 0.95]	estimated
beta_10_f	13.1029	4.77604	2.743	[0.05, 50]	estimated
beta_1_age_f	-0.122813	0.126048	-0.974	[-5, 5]	estimated
beta_1_age2_f	0.00730709	0.0102503	0.713	[-1, 1]	estimated
beta_1_nkids_f	-0.305471	1.14719	-0.266	[-5, 5]	estimated
theta_1_f	-1.68631	0.245846	-6.859	[-8, 0.95]	estimated
beta_E_m	-2.12326	0.319457	-6.646	[-25, 25]	estimated
beta_E_f	-2.81476	0.302078	-9.318	[-25, 25]	estimated
beta_h_pt1_m	-1.10936	0.379086	-2.926	[-10, 10]	estimated
beta_h_pt1_f	-0.517276	0.162932	-3.175	[-10, 10]	estimated
beta_h_pt2_m	0.0835202	0.237788	0.351	[-10, 10]	estimated
beta_h_pt2_f	1.15858	0.121438	9.541	[-10, 10]	estimated
beta_h_f35_m	2.2888	0.0821348	27.866	[-10, 10]	estimated
beta_h_f35_f	2.02835	0.0682392	29.724	[-10, 10]	estimated
beta_h_ft_m	2.38012	0.0865569	27.498	[-10, 10]	estimated
beta_h_ft_f	1.68251	0.0798241	21.078	[-10, 10]	estimated
beta_h_lh_m	0.69767	0.133487	5.226	[-10, 10]	estimated
beta_h_lh_f	-0.310873	0.161533	-1.925	[-10, 10]	estimated
beta_E_gsur	-1.19243	0.155667	-7.660	[-10, 10]	estimated
beta_E_drgn2	-0.154617	0.24423	-0.633	[-10, 10]	estimated
beta_E_drgn3	0.0872501	0.280833	0.311	[-10, 10]	estimated
beta_E_drgn4	0.0150136	0.306753	0.049	[-10, 10]	estimated
beta_E_drgn5	-0.145472	0.255453	-0.569	[-10, 10]	estimated
beta_E_drgn6	-0.284391	0.274814	-1.035	[-10, 10]	estimated
beta_E_drgn7	-0.139569	0.265133	-0.526	[-10, 10]	estimated
beta_E_drgn8	-0.165927	0.26603	-0.624	[-10, 10]	estimated
beta_E_drgur	-0.182322	0.165665	-1.101	[-10, 10]	estimated
beta_E_drgmd	-0.417356	0.1861	-2.243	[-10, 10]	estimated
beta_occ_2_m	-1.50975	0.0967347	-15.607	[-15, 15]	estimated
beta_occ_3_m	-2.25319	0.124405	-18.112	[-15, 15]	estimated
beta_occ_4_m	0.159368	0.061796	2.579	[-15, 15]	estimated

Coordinate	Estimate	CR1 s.e.	z	Box	Status
beta_occ_2_f	0.207379	0.0891054	2.327	[-15, 15]	estimated
beta_occ_3_f	-0.180381	0.0923137	-1.954	[-15, 15]	estimated
beta_occ_4_f	0.81589	0.079925	10.208	[-15, 15]	estimated
beta_w0	2.048	0.0335421	61.058	[-10, 20]	estimated
beta_w_educL	-0.0433046	0.0216643	-1.999	[-5, 5]	estimated
beta_w_educH	0.181652	0.0197059	9.218	[-5, 5]	estimated
beta_w_pexp	0.56435	0.0545928	10.337	[-3, 3]	estimated
beta_w_pexp2	-0.15983	0.0242888	-6.580	[-0.3, 0.3]	estimated
sigma	0.363075	0.0072235	50.263	[0.1, 20]	estimated
delta_occ_2	-0.0740322	0.0231318	-3.200	[-4, 4]	estimated
delta_occ_3	0.0393566	0.0228	1.726	[-4, 4]	estimated
delta_occ_4	0.214559	0.0223344	9.607	[-4, 4]	estimated
beta_c	2.10172	0.293879	7.152	[0.05, 50]	estimated

B Appendix B. Inequality indices and the allocation rule

The preliminary decomposition of Section 5 uses the Gini index only, on a weighted distribution of strictly positive money-metric levels with mean μ : $\mathcal{I} = \frac{1}{2\mu} \mathbb{E}|W - \tilde{W}|$, for $W, \tilde{W}$ independent draws from the distribution. Extending the exercise to further indices is future work and is not reported here.

The exact Shapley value on the three-player game $v(S) = I_\emptyset - I_S$, $S \subseteq \{P, A, B\}$, allocates to factor k the average of its marginal contribution $v(S \cup \{k\}) - v(S)$ over all $3! = 6$ orderings in which the three factors can be introduced:

$$C_k = \frac{1}{6} \sum_{\pi} [v(S_{\pi}(k) \cup \{k\}) - v(S_{\pi}(k))],$$

where $S_{\pi}(k)$ is the set of factors preceding k in ordering π . With three players this has a closed form and requires no grouping into unions. Contributions are signed and are never renormalised to sum to one hundred by construction: they sum to $\Delta I = I_\emptyset - I_{\{P,A,B\}}$ because the game closes, and the closure is verified.

C Appendix C. Data, software and replication

Data. French EU-SILC, accessed through Eurostat’s harmonised release, with 2016 survey collection and a 2015 income reference year. The harmonised survey is transformed into an input file for EUROMOD (Sutherland and Figari, 2013), which applies the French 2015 policy system. Regional labour-market conditions come from the Eurostat regional labour-force series. Access to EU-SILC microdata is granted by Eurostat under its research-access conditions and the data cannot be redistributed with this document.

Sample. 1,540 single-adult and 2,223 couple households, constructed by the screens of Section 2.

Estimator and inference. Conditional likelihood over 100 sampled alternatives per household plus the observed choice, with an out-of-fold proposal correction; cluster-robust standard errors on the household. Optimization is checked by five starts under two polishing contracts and curvature by exact Hessian eigenvalues.

Welfare computation. The accepted literal Mapping-F baseline is reported from its verified aggregate artifacts. The bounded DECOMP-2 exercise recomputes all eight coalitions in the three-player game for each reporting scale. Its uncertainty display is a Monte Carlo range across 1,000 simulation replications plus an independent second-seed check; it is not a parameter-uncertainty interval.

Software. Python with JAX (0.10.1) for automatic differentiation, and the EUROMOD connector (0.2.17).

Replication. Code and derived, non-confidential intermediate artefacts will be made available in a public repository on publication. Confidential microdata remain in their permitted environment, so the replication package reproduces every step conditional on authorised access to EU-SILC.

D Scientific history of this result

This section records how the reported specifications and results came to be what they are. It is here so that the argument above does not have to carry it, and so that a reader comparing this version with an earlier one can see what changed and why. Nothing in it is a competing model.

The consumption specification. Earlier drafts reported a specification in which the consumption coefficient was fixed at one as a numeraire, and one in which the single-adult consumption curvature was estimated at $\theta_c \approx 0.168$. Neither is the reported model. The current specification fixes the shock scale, fixes the consumption curvature at exactly zero, and estimates the consumption weight instead. That change improves the criterion by 23.753 log-points for single adults and 15.659 for couples, and it changes how a given non-consumption advantage converts into the money metric: under the closed form of Section 4, that advantage now enters divided by the estimated β_c rather than by the fixed numeraire it previously used. Every welfare quantity computed under the earlier convention is superseded, and the corresponding figures were regenerated rather than relabelled.

The sample. The predecessor frames were screened further when the hours support and occupation mapping were applied to observed rows, with a further exclusion where an observed job prices to non-positive disposable consumption. The current descriptive tables are computed on the resulting 1,540 and 2,223 households, not on the predecessor frames, and the funnel in Section 2 ends where the estimation begins.

The disposable-income convention. Single-adult disposable income was previously aggregated over the decider only; it is now aggregated over all resident household members, which is the convention the couples application always used. The two applications are now on the same accounting convention.

The decomposition result. Section 5 reports only the bounded, explicitly preliminary three-factor exercise: systematic utility heterogeneity, coarse geographic/temporal access shifters and earning opportunities, with resources, needs and composition held fixed. It is sourced from the DECOMP-2 replication run and reported with Monte Carlo simulation ranges and a second-seed check rather than parameter-uncertainty intervals. Earlier architectures and their numerical results are absent from every current reader-facing surface and remain only in the audit record.

The welfare reference. Section 4 reports only the accepted literal Mapping-F construction. Under the current specification the home reference carries no opportunity-density, proposal, shock or intensity argument. Superseded candidates and their numerical results are absent from every current reader-facing surface and remain only in the audit record.

The consumption normalizer. Three values of λ_c were in circulation because different panels recomputed it over their own rows. Under exact log consumption the constant is alternative-invariant and cancels from every choice probability; it does not enter the accepted W_F^1 welfare measure at all, so no welfare result is affected by which value is used. The paper now reports one constant per population, for the choice-probability role it actually plays.

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